

Reducing Dimensions of Semidefinite Programs with Wedderburn-Artin Theorem

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Key Reference: Vallentin, Frank. "Symmetry in semidefinite programs." *Linear Algebra and its Applications* 430.1 (2009): 360-369.

Semidefinite Programs

Symmetry of Semidefinite Programs

Simplification by Wedderburn-Artin Theorem

Example: Lovász Number ϑ of Cycle Graph

Complex Semidefinite Program

$$\begin{array}{ll}\max & \langle C, Y \rangle \\ \text{s.t.} & \langle A_i, Y \rangle = b_i, \quad i = 1, 2, \dots, n \\ & Y \succeq 0\end{array}$$

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- $Y \in \mathbb{C}^{X \times X}$ is a **variable** Hermitian matrix.
- $Y \succeq 0$ means that Y is **positive semidefinite**.
- $\langle C, Y \rangle = \sum_{i,j \in X} \overline{C_{ij}} Y_{ij} \in \mathbb{R}$ since C, Y are Hermitian.

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- We assume that this program has an optimal solution.

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- Semidefinite programming (SDP) has a wide range of applications:
 - ▶ combinatorial optimization, control theory...
- The dimensions of matrices causes difficulty for solving SDP.
 - ▶ It is crucial to exploit problems' structure to reduce the dimension.
- We introduce tools in representation theory to reduce the dimension of SDP when the problem has symmetry.
 - ▶ First we define the **symmetry** of SDP problems.
 - ▶ Second we explain how **Wedderburn-Artin theorem** reduces the dimension.
 - ▶ Finally we give an example of such simplification.

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Symmetry of Semidefinite Programs

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 - ▶ For $a \in G, (x, y) \in X \times X$, define $a(x, y) := (ax, ay)$.
- The group action extends to $X \times X$ -matrix:
 - ▶ For an $X \times X$ -matrix M and $x, y \in X$, define $aM(x, y) := M(ax, ay)$.
 - ▶ M is **invariant under G** if $M = aM$ for all $a \in G$.

Symmetry of SDP

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- Such optimal solution is invariant under G .

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 - ▶ $\mathcal{B}$ forms a vector space over $\mathbb{C}$.
 - ▶ In fact, $\mathcal{B}$ is a **semisimple algebra**, which will be shown in next section.
- As discussed before, the SDP invariant under G always has an optimal solution invariant under G .
- It is equivalent to solve:

$$\begin{aligned} \max \quad & \langle C, Y \rangle \\ \text{s.t.} \quad & \langle A_i, Y \rangle = b_i, \quad i = 1, 2, \dots, n \\ & Y \succeq 0, \quad Y \in \mathcal{B} \end{aligned}$$

- By the action of G , $X \times X$ can be partitioned into the orbits: $R_1, R_2, \dots, R_N$.

Canonical Basis for $\mathcal{B}$

- By the action of G , $X \times X$ can be partitioned into the orbits: $R_1, R_2, \dots, R_N$.
- For every R_r , we define the following matrix:

$$B_r \in \{0, 1\}^{X \times X} \quad B_r(x, y) = \begin{cases} 1 & \text{if } (x, y) \in R_r \\ 0 & \text{otherwise.} \end{cases}$$

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 - ▶ It is clear that $B_1, B_2, \dots, B_N$ are linearly independent.
 - ▶ For each $M \in \mathcal{B}$ and each R_r , since M is invariant under G , we have

$$M(x, y) = M(x', y') \quad \forall (x, y), (x', y') \in R_r$$

First Step of Simplification

$B_1, B_2, \dots, B_N$ forms a basis for $\mathcal{B}$

■ For each $Y \in \mathcal{B}$, we have $Y = y_1 B_1 + \dots + y_N B_N$ with $y_i \in \mathbb{C}$.

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► To make Y Hermitian, we ask $y_i = \bar{y}_j$ if $B_i = B_j^\top$.

■ Let $c_r = \langle C, B_r \rangle$ and $a_{ir} = \langle A_i, B_r \rangle$, the SDP can be reformulated as follows:

$$\max \quad c_1 y_1 + \dots + c_N y_N$$

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$$a_{i1} y_1 + \dots + a_{iN} y_N = b_i, \quad i = 1, \dots, n$$

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- In application, the dimension of $B_r \in \{0, 1\}^{X \times X}$ could be large.
- For each $B_r \in \mathcal{B}$, we aim to reduce its dimension:
 - ▶ First, we show that $\mathcal{B}$ is a **semisimple algebra**.
 - ▶ Second, we use **Wedderburn-Artin theorem** for simplification.

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- This implies that $M = 0$. Contradicts.

Wedderburn-Artin Theorem

Theorem (Wedderburn-Artin)

Let A be a finite-dimensional semisimple algebra over a field F . Then,

$$A \cong M_{n_1}(D_1) \times M_{n_2}(D_2) \times \cdots \times M_{n_k}(D_k)$$

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If the base field F is algebraically closed (such as $\mathbb{C}$), there are no finite-dimensional division algebras over F other than F itself. Therefore,

$$A \cong M_{n_1}(F) \times M_{n_2}(F) \times \cdots \times M_{n_k}(F)$$

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$$\varphi : \mathcal{B} \rightarrow \bigoplus_{i=1}^k M_{n_i}(\mathbb{C})$$

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- In many applications, $\sum_{i=1}^k n_i \ll |X|$.

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- There is an explicit construction of φ .
 - ▶ Such construction can be turned into an algorithm.
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 - ▶ Such construction can be turned into an algorithm.
 - ▶ Symbolic calculation by hand is also possible.
- The process of construction leverages character theory and Maschke's theorem.
- We are going to use an example to demonstrate the simplification briefly.

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Example: Lovász Number ϑ of Cycle Graph

SDP Formulation for Lovász Number ϑ of C_n

- Consider the cycle graph $C_n = (V, E)$.

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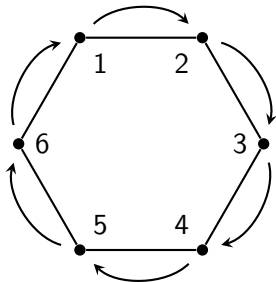
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- Here Y is a $V \times V$ matrix. When $|V|$ is large, solving this SDP is time-consuming.

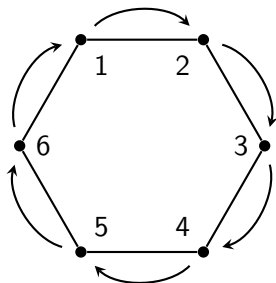
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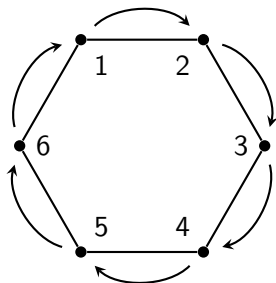


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- It is clear that the SDP is invariant under G .

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▶ Y must be a circulant matrix (left figure).

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■ We can construct a unitary matrix F such that

► $\varphi(Y) = F^* Y F = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_{n-1}) \quad \forall Y \in \mathcal{B}.$

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